

Chang Xu

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Education

PhD	The Hong Kong University of Science and Technology , Physics	HK SAR Aug 2022 – present
BSc	University of Science and Technology of China , Applied Physics	Hefei, China Aug 2018 – July 2022

Experience

Department of Physics, The Hong Kong University of Science and Technology , PhD Student	HKUST, HK SAR Aug 2022 – present 4 years
Working with Prof. Qin Xu to design the nonlinear mechanics of soft composite solids.	
Department of Modern Physics, University of Science and Technology of China , Student Research Assistant	USTC, China June 2020 – July 2022 2 years 2 months
Working with Prof. Bing-Hong Wang to investigate the asymmetric simple exclusion process on complex networks.	

Presentations

- CRF Joint Meeting on Active Matter at HKUST (Hong Kong SAR, 2025-11).
- 2025 APS March Meeting (Anaheim, USA, 2025-03).
- 2024 USTC-HKUST Joint Workshop on Condensed Matter Physics (Chaohu, China, 2024-12).
- 2024 Frontiers in Non-equilibrium Physics (Kyoto, Japan, 2024-07).
- 2023 USTC-HKUST Joint Workshop on Condensed Matter Physics (Guangzhou, China, 2023-12).

Skills

Design, construct and conduct mechanical experiments: CAD; 3D printing; nano-indentation; Raspberry Pi/Arduino automation; spin coating; metal/plastic processing with milling/drilling machine; bench lathe and other machine shop tools.

Imaging soft and complex systems using advanced techniques: Traction force microscopy, confocal microscopy, interferometry, and photoelasticimetry.

Digital image processing: Object identification, tracking, and stress field reconstructions.

Numerical simulation and modelling: Molecular Dynamics and Monte Carlo simulations.

Research Interests

Interests: Soft materials, Granular physics, Active matter, Biophysics, Biomechanics, Mechanobiology, Statistical mechanics, Nonlinear dynamics, Stochastic processes